

Multi-omics fingerprint of *in vitro* bioengineered heart-on-chip platform

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INTRODUCTION

In the last decades, there has been increased demand to clearly define the effects of pollutants in ecosystems and living organisms, especially those without regulation. Contrarily to this raising of environmental awareness, animal testing addresses chemical toxicity. To avoid this particular paradox and to incorporate the 3R's principle in toxicology, tissue engineering is being used to develop more innovative approaches to test toxicological effects and identify the risk assessment of potentially harmful chemicals present in the environment. The use of different scaffolds, materials and gel-based matrices to grow and maintain the cells are the main cores of tissue bioengineering.

However, although some determinations in these systems are already developed and routinely analyzed (cell viability, oxidative stress...), any multi-omics approach is conducted, and only scarce applications on individual omics analysis (some proteomics studies or targeted transcriptomics by PCR) can be found. Two main problems can be encountered when facing omics analysis using organ-on-chip: 1) the obtention of the embedded cells in the matrix maintaining intact their integrity to, afterwards, extract the molecules of interest; and 2) the limited amount of sample to perform different omics analysis (especially critical in metabolomics).

Here, we developed and optimized the protocols to analyze, under a multi-omics perspective (metabolomics, proteomics and transcriptomics), samples derived from a gelatin methacryloyl-based matrix (GelMA) for a New Approach Methodology (NAM) platform in the assessment of toxicology without the use of animal models. The optimization was performed not only for the analytical methodologies, but also for the cell harvesting and the extraction protocols to obtain the maximum coverage of metabolites, proteins and RNA.

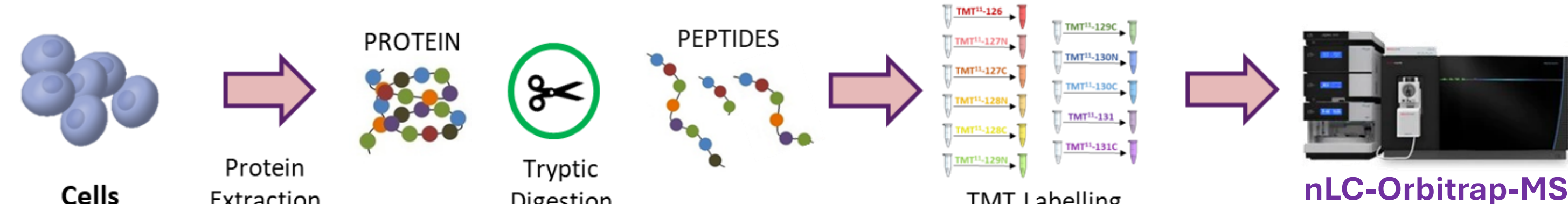
MATERIALS AND METHODS



Metabolomics



Proteomics



Transcriptomics



RESULTS

These methodologies detected around 20000 genes, 500 proteins and 200 metabolites involved in glycolysis, Krebs cycle, biosynthesis of amino acids, lipogenesis/lipolysis, calcium signalling pathway, cardiac contraction and cell adhesion.



The proposed protocol is a fast and easy methodology to perform a multi-omics approach in 3D cultured cells specifically designed for a bioreactor.

The enzymatic method based on collagenase was the most suitable for obtaining the highest yield and coverage of molecules to perform metabolomics, proteomics and transcriptomics analyses with a low amount of sample.

